10. Given that $f(t) \star g(t) = y(t)$, where $\star$ denotes convolution,

(a) Find $f(t - T_1) \star g(t - T_2)$, for some finite-valued real numbers T_1 and T_2.

(b) Use the result of (a) and the fact that $u(t) \star u(t) = r(t)$, to find $(u(t+1) - u(t-2)) \star (u(t-3) - u(t-4))$. Verify the result graphically.

(a)

$$p(t) = \int_{-\infty}^{\infty} f(\tau - T_1)\, g(t - \tau - T_2)\, d\tau$$

$$\tau_1 = \tau - T_1$$

$$= \int_{-\infty}^{\infty} f(\tau_1)\, g(t - \tau_1 - (T_1 + T_2))\, d\tau_1$$

$$= y(t - (T_1 + T_2))$$

(b) $[u(t+1) - u(t-2)] * [u(t-3) - u(t-4)]$

$$= [u(t+1) * u(t-3)] - [u(t+1) * u(t-4)] - [u(t-2) * u(t-3)] + [u(t-2) * u(t-4)]$$

$$= r(t-2) - r(t-3) + r(t-6) - r(t-5)$$